

MOHAMMED SEDEG

Senior AI Engineer · LLM Systems & Multi-Agent · Multimodal Deep Learning · MLOps

Turkey (open to relocate) · silvapi1994@gmail.com · [LinkedIn](#) · [GitHub](#) · [Portfolio](#) · [Certificates & Courses](#)

PROFESSIONAL SUMMARY

AI Engineer with 5+ years shipping production LLM systems and edge computer vision, backed by 9 years across automation and renewable energy. Engaged directly by clients for senior, end-to-end ownership since 2023: cut a fintech KYC pipeline's cost from **~\$100K/month to ~\$30/month** (>99.9%) while fixing a critical data-compliance risk, delivered **3× faster LLM inference** via GGUF/ExLlamaV2, cut fine-tuning cost **70%** via LoRA/QLoRA, and scaled computer vision to **30 concurrent camera streams** in production. Ships multi-agent LLM pipelines end-to-end — architecture, evals, observability.

CORE COMPETENCIES

LLMs & Fine-Tuning Hugging Face · LangChain · LoRA / QLoRA · RLHF / DPO · Multi-agent pipelines · LLM evals	RAG & Vector Search pgvector · Qdrant · Weaviate · FAISS · ChromaDB · MongoDB · Redis · Hybrid retrieval · Re-ranking
Multimodal & Vision LLaVA · Qwen-VL · YOLOv8 · NVIDIA DeepStream · ONNX / TFLite · OpenCV · Detectron2 · Jetson / Raspberry Pi	Training & Optimization DeepSpeed · PyTorch DDP / FSDP · AMP · GGUF · ExLlamaV2 · TensorRT
MLOps & Infrastructure Docker · FastAPI · AWS SageMaker · MLflow · Prometheus · Grafana · Prefect · GitHub Actions	ML & Research PyTorch · TensorFlow · JAX · scikit-learn · XGBoost · LSTM · Transformer · CNN
Languages Python (advanced) · C/C++ · R · Bash · SQL	Data & Storage PostgreSQL · SQLAlchemy · PySpark · Alembic

PROFESSIONAL EXPERIENCE

- Senior AI Engineer & LLM Consultant · Independent / Contract2023 – Present
- Fintech KYC Document Intelligence — Cost & Compliance Redesign:** Initial pipeline brute-forced document OCR/extraction through Gemini Vision for every submission — functionally correct, but projected at **~\$100K/month** and, more critically, sent raw user PII to a third-party API with no compliance guardrails. Redesigned as a self-hosted **complexity-classification router**: simple, high-confidence documents resolve through a lightweight in-house image-processing path, while only ambiguous documents escalate to a self-hosted Mistral OCR + structured-extraction model — cutting cost to **~\$30/month** and keeping all client data on-premises; cut manual processing **60%** and improved extraction reasoning **20%** via Chain-of-Thought/Tree-of-Thought prompting.
 - Enterprise RAG Architecture:** Shipped multi-provider LLM systems (OpenAI, Cohere, Hugging Face, Ollama) with modular retriever-generator pipelines over pgvector, Qdrant, and MongoDB via FastAPI — reducing query latency **40%** on production workloads.
 - LLM Fine-Tuning at Scale:** Cut client fine-tuning costs **70%** via LoRA and QLoRA on custom instruction datasets across 7B–13B parameter models; accelerated distributed training 20–30% using DeepSpeed, PyTorch DDP, and Automatic Mixed Precision.
 - Inference Optimization:** Deployed GGUF and ExLlamaV2 quantized models achieving **3× faster inference** at under 50% memory footprint; built full observability stack (Prometheus + Grafana) and orchestrated pipelines with Prefect — cutting deployment cycle 35%.
- AI Engineer — Computer Vision Specialist · VisionCore2024 – 2025
- Edge AI Deployment:** Shipped low-latency TFLite and ONNX models on <4 GB RAM hardware achieving 60–80 ms end-to-end inference for real-time industrial applications.
 - PPE Detection System at Scale:** Developed a YOLOv8-based workplace-safety detector reaching **85–90% mAP** across 8 PPE categories; deployed via **NVIDIA DeepStream** as a multi-stream inference pipeline processing **30**

concurrent camera feeds on a **4×RTX 4070 GPU server**; reduced overfitting 15% through advanced data augmentation (Mosaic, MixUp, Albumentations).

- **MLOps Automation:** Automated annotation and augmentation workflows via Roboflow and VLM-assisted labeling — cutting manual effort 25% and doubling dataset iteration speed.

R&D Engineer — AI & Deep Learning · Karabuk University

2021 – 2023

- **Multimodal Solar Forecasting:** Designed an LSTM-Transformer-CNN fusion architecture over NOAA GOES-16/17 satellite imagery with cross-modal attention — outperforming published baselines on RMSE, MAE, and R². Manuscript currently under peer review.
- **V2G Microgrid Optimization:** Built a Vehicle-to-Grid optimization pipeline using Improved Antlion Optimizer and PSO with Monte Carlo simulation — delivering full techno-economic benchmarking (COE, NPC, LPSP, REF) for renewable microgrid configurations.

Earlier Engineering Roles · Industrial Automation & Renewable Energy

2017 – 2020

- PLC/HMI/SCADA control systems (uptime +18%), solar deployments including a dual-axis tracker (+35% energy capture), and hybrid solar-EV R&D from concept to prototype.

LEADERSHIP & COMMUNITY

SAiR — Sudanese AI Research Initiative · Founder & Lead Instructor

- Founded and lead a free, open-source AI engineering program — a **500+ member** community reaching **300+ active learners** across Sudan and Africa with a **65% completion rate**; graduates report **94% confidence** in technical interviews and land offers in an average of **3.2 months**.
- Built and teach a 6-module curriculum (Python/data science → ML → deep learning → GPT-from-scratch → MLOps). Flagship capstone **miniGPT** is a from-scratch GPT-2 (124M-param) implementation — tokenizer, multi-head attention, DDP training loop, cloud training on Modal A100, web UI — shipped with 39 passing tests; student teams have gone on to ship production systems in healthcare diagnostics, crop-disease detection, and Arabic NLP.

KEY PROJECTS

MyLLM — Transparent From-Scratch LLM Framework

- Installable pure-PyTorch framework covering the full LLM pipeline (tokenization → attention → training → RLHF → inference) with first-class support for GPT-2, LLaMA-2, and LLaMA-3, one-line `LLM.from_pretrained()`, configurable SFT/DPO/PPO trainers, and single-GPU/DDP/DeepSpeed/FSDP accelerators.
- Backed by 128 passing tests (CPU-only, ~15s), 21 learning notebooks, 5 Colab demos, and 34-page documentation.

RagApp — End-to-End Retrieval-Augmented Generation System (v2)

- Production-ready async RAG platform on FastAPI with dual vector-DB support (PostgreSQL + pgvector primary, Qdrant optional) switchable via a single env variable, custom chunking, multi-provider embeddings/LLMs, and Dockerized services.

Fero Cafe (Cafe.AI) — Multi-Agent Conversational Commerce System

- Full-stack 5-agent async pipeline (Guard → Classification → Details/Order/Recommendation) with RAG over ChromaDB, SSE streaming, and a live observability dashboard; FastAPI + React Native, powered by Llama-3.3-70B via Groq. 160 passing tests, LLM evals (80% pass-rate gating), CI/CD via GitHub Actions.

V2G Microgrid Optimization & Solar Forecasting — Research Framework

- Python framework pairing V2G microgrid optimization (Improved Antlion Optimizer + PSO + Monte Carlo) with a multimodal LSTM-Transformer-CNN solar-irradiance forecasting system.

EDUCATION

M.Sc. Mechatronics — Specialization: Computer Vision · 2023

Karabuk University, Turkey

B.Sc. Electrical Engineering — Specialization: Control Systems · 2016

Sudan University of Science and Technology

LANGUAGES

Arabic — Native · English — Fluent (C1) · Turkish — Intermediate (B2)